

Probabilistic Graphical models

Sunita Sarawagi
IIT Bombay

<http://www.cse.iitb.ac.in/~sunita>

Probabilistic modeling

- Given: several variables: $x_1, \dots, x_n$, n is large.
- Task: build a joint distribution function $\Pr(x_1, \dots, x_n)$
- Goal: Efficiently represent, estimate, and answer inference queries on the distribution
- Basic premise
 - ▶ Explicit joint distribution is dauntingly large
 - ▶ Queries are simple **marginals** (sum or max) over the joint distribution.

Example

- Variables are attributes are people.

Age	Income	Experience	Degree	Location
10 ranges	70 scales	70 scales	3 scales	3000 places

- An explicit joint distribution over all columns not tractable:
number of combinations:
 $10 \times 70 \times 70 \times 3 \times 3000 = 441,000,000.$

Alternatives to an explicit joint distribution

- Assume all columns are independent of each other: **bad assumption**
- Use data to detect pairs of highly correlated column pairs and estimate their pairwise frequencies
 - ▶ Many highly correlated pairs
income $\not\perp$ age, income $\not\perp$ experience, age $\not\perp$ experience
 - ▶ **Ad hoc methods of combining these into a single estimate**
- Go beyond pairwise correlations: conditional independencies
 - ▶ income $\not\perp$ age, but income $\perp$ age | experience
 - ▶ experience $\perp$ degree, but experience $\not\perp$ degree | income

Graphical models make explicit an efficient joint distribution from these independencies

More examples of Conditional Independencies (CIs)

- The grades of a student in various courses are correlated but they become CI given attributes of the student (hard-working, intelligent, etc?)
- Health symptoms of a person may be correlated but are CI given the latent disease.
- Words in a document are correlated, but may become CI given the topic. $x_1, x_2, x_{i-2}, x_{i-1}, x_i, x_n$
- Pixel color in an image become CI of distant pixels given near-by pixels.



Graphical models

Model joint distribution over several variables as a product of smaller factors that is

- 1 Intuitive to represent and visualize
 - ▶ Graph: represent structure of dependencies
 - ▶ Potentials over subsets: quantify the dependencies
- 2 Efficient to query
 - ▶ given values of any variable subset, reason about probability distribution of others.
 - ▶ many efficient exact and approximate inference algorithms

Graphical models = graph theory + probability theory.

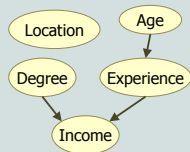
Representation

Structure of a graphical model: Graph + Potential

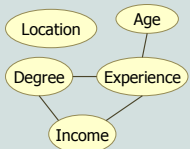
Graph

- Nodes: variables $\mathbf{x} = x_1, \dots, x_n$
 - ▶ Continuous: Sensor temperatures, income
 - ▶ Discrete: Degree (one of Bachelors, Masters, PhD), Levels of age, Labels of words
- Edges: direct interaction
 - ▶ Directed edges: Bayesian networks
 - ▶ Undirected edges: Markov Random fields

Directed



Undirected



Representation

Potentials: $\psi_c(\mathbf{x}_c)$

- Scores for assignment of values to subsets c of directly interacting variables.
- Which subsets? What do the potentials mean?
 - ▶ Different for directed and undirected graphs

Probability

Factorizes as product of potentials

$$\Pr(\mathbf{x} = x_1, \dots, x_n) \propto \prod \psi_S(\mathbf{x}_S)$$

Directed graphical models: Bayesian networks

- Graph G : directed acyclic
 - ▶ Parents of a node: $\text{Pa}(x_i)$ = set of nodes in G pointing to x_i
- Potentials: defined at each node in terms of its parents.

$$\underline{\psi_i(x_i, \text{Pa}(x_i)) = \Pr(x_i | \text{Pa}(x_i))}$$

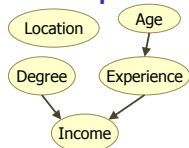
$\text{Pa}(x_3) = \{x_1, x_2\}$



- Probability distribution

$$\underline{\Pr(x_1 \dots x_n) = \prod_{i=1}^n \Pr(x_i | \text{pa}(x_i))}$$

Example of a directed graph



$$\psi_1(L) = \Pr(L)$$

NY	CA	London	Other
0.2	0.3	0.1	0.4

$$\psi_2(A) = \Pr(A)$$

20-30	30-45	> 45
0.3	0.4	0.3

or, a Gaussian distribution
 $(\mu, \sigma) = (35, 10)$

$$\psi_3(E, A) = \Pr(E|A)$$

	0-10	10-15	> 15
20-30	0.9	0.1	0
30-45	0.4	0.5	0.1
> 45	0.1	0.1	0.8

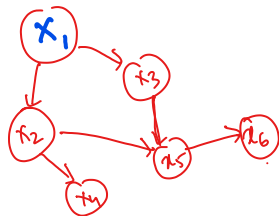
$$\psi_5(I, E, D) = \Pr(I|D, E)$$

3 dimensional table, or a
 histogram approximation.

Probability distribution

$$\text{Pa}(\mathbf{x} = L, D, I, A, E) = \Pr(L) \Pr(D) \Pr(A) \Pr(E|A) \Pr(I|D, E)$$

Definitions on A DAG



- 1 Descendants of a node $D(x_i)$: set of nodes that are reachable from it via directed paths. $D(x_3) = \{x_5, x_6\}$
- 2 Non-descendants $ND(x_i)$: The remaining nodes. That is, nodes that cannot be reached from x_i . $ND(x_3) = \{x_1, x_2, x_4\}$
- 3 Spouse of a node $Sp(x_i)$: Nodes with which x_i shares a child. $Sp(x_3) = x_2$

Conditional Dependencies

- Given three sets of variables X, Y, Z , set X is conditionally independent of Y given Z ($X \perp\!\!\!\perp Y | Z$) iff

$$\Pr(X|Y, Z) = \Pr(X|Z)$$

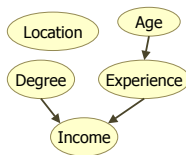
$$\Pr(G_{191} | G_{125}, H, I) = \Pr(G_{191} | H, I)$$

$G_{191} \perp\!\!\!\perp G_{125} | \{H, I\}$
 $\underbrace{\quad}_X \quad \underbrace{\quad}_Y \quad \underbrace{\quad}_Z$

- Local conditional independencies in BN: for each x_i

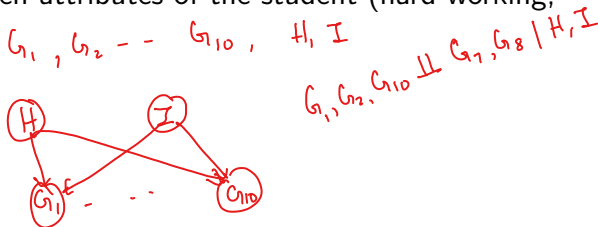
$$\underline{x_i} \perp\!\!\!\perp \underline{ND(x_i)} | Pa(x_i)$$

- $L \perp\!\!\!\perp E, D, A, I$
- $A \perp\!\!\!\perp L, D$
- $E \perp\!\!\!\perp L, D | A$
- $I \perp\!\!\!\perp A, L | E, D$



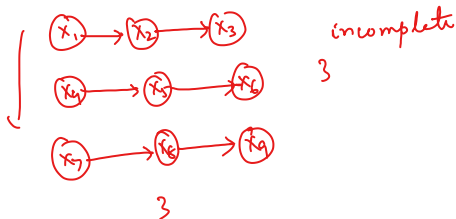
More examples of BNs representing Conditional Independencies (CIs) I

- The grades of a student in various courses are correlated but they become CI given attributes of the student (hard-working, intelligent, etc?)



More examples of BNs representing Conditional Independencies (CIs) II

- Pixel color in an image become CI of distant pixels given near-by pixels.



Drawing a BN starting from a distribution

Given a distribution $P(x_1, \dots, x_n)$ to which we can ask any CI of the form "Is $X \perp\!\!\!\perp Y | Z$?" and get a yes/no answer.

Goal: Draw a minimal, correct BN G to represent P .

- A DAG G is correct if all Local-CIs that are implied in G hold in P .
- A DAG G is minimal if we cannot remove any edge(s) from G and still get a correct BN for P .

$$V \equiv \{x_1, \dots, x_n\}$$

for every x_i find a S_i s.t. $x_i \perp\!\!\!\perp \boxed{V - S_i} \mid S_i$

$$x_1 \perp\!\!\!\perp \underline{x_2 \dots x_n} \mid \underline{0}$$

not necessarily the $ND(x_i)$

Algorithm for drawing a BN from CIs

$x_1, \dots, x_n$ = Choose an ordering of variables

For $i = 1 \dots n$

- S = smallest subset of $Q_i = \{x_1, \dots, x_{i-1}\}$ such that

$$\underline{x_i \perp\!\!\!\perp Q_i - S \mid S}$$

- Make each variable in S a parent of x_i

Examples

G_1, G_2, H, I

Initial order: $\{H, G_1, I, G_2\}$ $i=1$

$i=2$ $G_1 \perp\!\!\!\perp H - S \mid S$ $S \subset \{H\}$

$S = \emptyset$ $G_1 \perp\!\!\!\perp H$ $\times$

$S = H$ $G_1 \perp\!\!\!\perp \emptyset \mid H$ $\checkmark$

$i=3$ $I \perp\!\!\!\perp \{H, G_1\} - S \mid S$

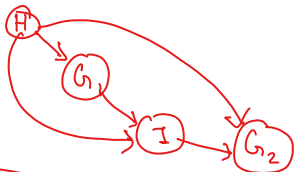
$S = \emptyset$ $I \perp\!\!\!\perp \{H, G_1\}$ $\times$

$S = H$ $I \perp\!\!\!\perp G_1 \mid H$ $\times$

$S = G_1$ $I \perp\!\!\!\perp H \mid G_1$ $\times$

$S = H, G_1$

G



$i=4$

G_2 :